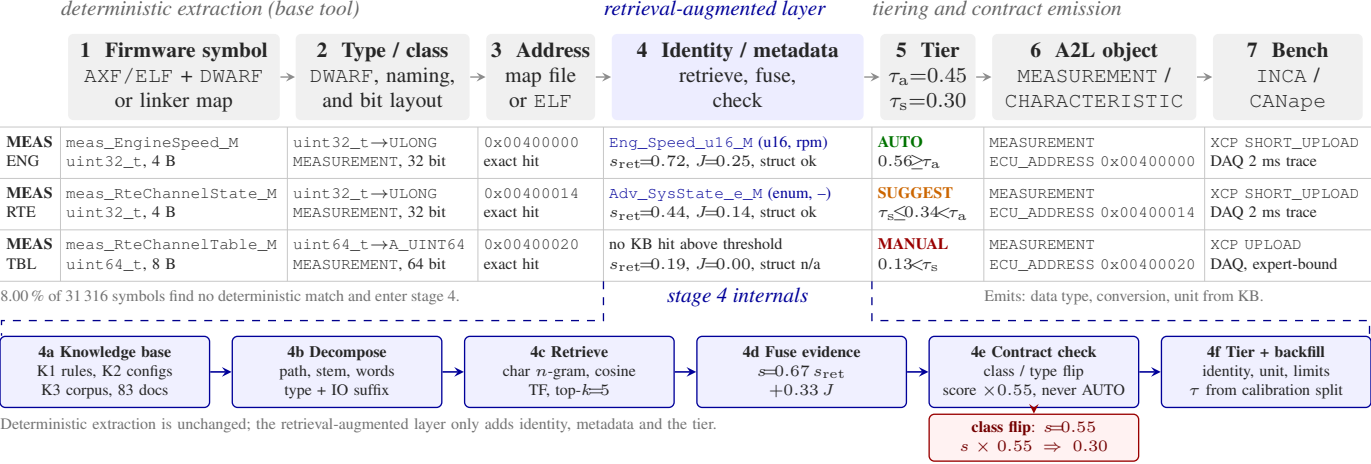
deterministic extraction (base tool)
retrieval-augmented layer
tiering and contract emission
1 Firmware symbol
AXF/ELF + DWARF
or linker map
2 Type / class
DWARF, naming,
and bit layout
3 Address
map file
or ELF
4 Identity / metadata
retrieve, fuse,
check
5 Tier
τ_a=0.45
τ_s=0.30
6 A2L object
MEASUREMENT /
CHARACTERISTIC
7 Bench
INCA /
CANape
MEAS ENG
meas_EngineSpeed_M
uint32_t, 4 B
uint32_t→ULONG
MEASUREMENT, 32 bit
0x00400000
exact hit
Eng_Speed_u16_M (u16, rpm)
s_{ret}=0.72, J=0.25, struct ok
AUTO
0.56≥τ_a
MEASUREMENT
ECU_ADDRESS 0x00400000
XCP SHORT_UPLOAD
DAQ 2 ms trace
MEAS RTE
meas_RteChannelState_M
uint32_t, 4 B
uint32_t→ULONG
MEASUREMENT, 32 bit
0x00400014
exact hit
Adv_SysState_e_M (enum, –)
s_{ret}=0.44, J=0.14, struct ok
SUGGEST
τ_s≤0.34<τ_a
MEASUREMENT
ECU_ADDRESS 0x00400014
XCP SHORT_UPLOAD
DAQ 2 ms trace
MEAS TBL
meas_RteChannelTable_M
uint64_t, 8 B
uint64_t→A_UINT64
MEASUREMENT, 64 bit
0x00400020
exact hit
no KB hit above threshold
s_{ret}=0.19, J=0.00, struct n/a
MANUAL
0.13<τ_s
MEASUREMENT
ECU_ADDRESS 0x00400020
XCP UPLOAD
DAQ, expert-bound
8.00 % of 31 316 symbols find no deterministic match and enter stage 4.
stage 4 internals
Emits: data type, conversion, unit from KB.
4a Knowledge base
K1 rules, K2 configs
K3 corpus, 83 docs
4b Decompose
path, stem, words
type + IO suffix
4c Retrieve
char n-gram, cosine
TF, top-k=5
4d Fuse evidence
s=0.67 s_{ret}
+0.33 J
4e Contract check
class / type flip
score ×0.55, never AUTO
4f Tier + backfill
identity, unit, limits
τ from calibration split
class flip: s=0.55
s × 0.55 ⇒ 0.30
Deterministic extraction is unchanged; the retrieval-augmented layer only adds identity, metadata and the tier.